

WARM-UP QUESTIONS FROM KREYZIG

Homogenous:

Case	Roots of (2)	Basis of (1)	General Solution of (1)
I	Distinct real λ_1, λ_2	$e^{\lambda_1 x}, e^{\lambda_2 x}$	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
II	Real double root $\lambda = -\frac{1}{2}a$	$e^{-ax/2}, xe^{-ax/2}$	$y = (c_1 + c_2 x)e^{-ax/2}$
III	Complex conjugate $\lambda_1 = -\frac{1}{2}a + i\omega,$ $\lambda_2 = -\frac{1}{2}a - i\omega$	$e^{-ax/2} \cos \omega x$ $e^{-ax/2} \sin \omega x$	$y = e^{-ax/2}(A \cos \omega x + B \sin \omega x)$

Find general solutions:

$$4y'' - 25y = 0$$

$$y'' + 36y = 0$$

$$100y'' + 240y' + (196\pi^2 + 144)y = 0$$

$$9y'' - 30y' + 25y = 0$$

Solve the IVPs:

$$y'' + 25y = 0, \quad y(0) = 4.6, \quad y'(0) = -1.2$$

$$y'' + y' - 6y = 0, \quad y(0) = 10, \quad y'(0) = 0$$

$$4y'' - 4y' - 3y = 0, \quad y(-2) = e, \quad y'(-2) = -e/2$$

$$y'' - y = 0, \quad y(0) = 2, \quad y'(0) = -2$$

$$y'' - k^2 y = 0 \quad (k \neq 0), \quad y(0) = 1, \quad y'(0) = 1$$

Now try these:

$$Y'' + 10Y' + 25Y = 0, \quad Y(0) = 3, \quad Y'(0) = 7$$

$$y'' + 4y' + 8y = 0, \quad y(0) = 2, \quad y'(0) = 6$$

$$y'' + 6y' + 13y = 0, \quad y(0) = 1, \quad y'(0) = 5$$

Non-homogeneous:

Table 2.1 Method of Undetermined Coefficients

Term in $r(x)$	Choice for $y_p(x)$
$ke^{\gamma x}$	$Ce^{\gamma x}$
kx^n ($n = 0, 1, \dots$)	$K_n x^n + K_{n-1} x^{n-1} + \dots + K_1 x + K_0$
$k \cos \omega x$	$\left\{ K \cos \omega x + M \sin \omega x \right.$
$k \sin \omega x$	
$ke^{\alpha x} \cos \omega x$	$\left\{ e^{\alpha x} (K \cos \omega x + M \sin \omega x) \right.$
$ke^{\alpha x} \sin \omega x$	

$$10y'' + 50y' + 57.6y = \cos x$$

$$y'' + 3y' + 2y = 12x^2$$

$$y'' - 9y = 18 \cos \pi x$$

$$y'' + 4y' + 4y = e^{-x} \cos x$$

Solve IVPs:

$$y'' + 3y = 18x^2, \quad y(0) = -3, \quad y'(0) = 0$$

$$y'' + 4y = -12 \sin 2x, \quad y(0) = 1.8, \quad y'(0) = 5.0$$

$$8y'' - 6y' + y = 6 \cosh x, \quad y(0) = 0.2, \\ y'(0) = 0.05$$

Now solve THIS:

$$y'' + 4y' + 4y = 5e^{-2x} + 12x^2 + 8e^{-2x} \cos x.$$